ALTERNATING SERIES

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Name: Solutions

Theorem. *Let $\{b_n\}$ be a sequence with positive terms, $0 < b_n$. If there exists some N such that*

(1) $b_{n+1} \leq b_n$ *whenever* $n \leq N$ *and*

(2) $\lim_{n\to\infty} b_n = 0$

then the Alternating Series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n$$

converges.

Decide whether the following series converge or diverge.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{3+5n}$ Converges by A.S.T.:

$f(x) = \frac{1}{3+5x}$, $f'(x) = \frac{-5}{(3+5x)^2} < 0$ for all $x \Rightarrow b_{n+1} = f(n+1) \leq f(n) = b_n$

$f(x) \to 0$ as $x \to \infty \Rightarrow b_n \to 0$ as $n \to \infty$

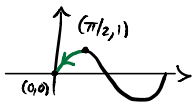
2. $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{3n-1}{2n+1}$ Diverges by the nth Term Test for Divergence.

Diverges: $\frac{3n-1}{2n+1} \to \frac{3}{2} \neq 0$, so

$\displaystyle\lim_{n\to\infty} (-1)^n \frac{3n-1}{2n+1}$ does not exist.

3. $\sum_{n=1}^{\infty}(-1)^n \sin\left(\frac{\pi}{n}\right)$ Converges by the A.S.T.:

$0 < \frac{\pi}{n} \leq \pi$ for all n, so when $n \geq 2$ $\sin\left(\frac{\pi}{n+1}\right) \leq \sin\left(\frac{\pi}{n}\right)$:



and $\lim_{n\to\infty} \sin\left(\frac{\pi}{n}\right) = \sin(0) = 0$

4. $\sum_{n=1}^{\infty}(-1)^n \frac{n^2}{n^2+n+1}$ Diverges by the nth Term Test for Divergence:

$$\lim_{n\to\infty} \frac{n^2}{n^2+n+1} = 1 \Rightarrow \lim_{n\to\infty} (-1)^n \frac{n^2}{n^2+n+1} \text{ Does Not Exist.}$$

5. For what values of p is the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$$

convergent?

$f(x) = \frac{1}{x^p} \Rightarrow f'(x) = \frac{d}{dx} x^{-p} = -p x^{-p-1} = \frac{-p}{x^{p+1}} < 0$ for all $x > 0$

$\Rightarrow \frac{1}{(n+1)^p} = f(n+1) \leq f(n) = \frac{1}{n^p}$

and

$$\lim_{n\to\infty} \frac{1}{n^p} = \lim_{x\to\infty} f(x) = \left(\frac{1}{x}\right)^p = \begin{cases} 0 & p>0 \\ 1 & p=0 \\ \infty & p<0 \end{cases}$$

so the series converges when $p>0$ by the A.S.T. and diverges for $p \leq 0$ by the nth term test for divergence.

6. Approximate the sum of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6}$$

correct to four decimal places.

We need

$$|R_n| \leq \frac{1}{(n+1)^6} < \frac{1}{10^4} \Leftrightarrow 10^4 < (n+1)^6 \Leftrightarrow 10^{4/6} = 10^{2/3} < n+1 \Leftrightarrow 10^{2/3} \leq n$$

Since

$$4^3 = 64 < 100 = 10^2 < 125 = 5^3$$

we have

$$4 < 10^{2/3} < 5$$

so

$$S_5 = \frac{1}{1^6} - \frac{1}{2^6} + \frac{1}{3^6} - \frac{1}{4^6} + \frac{1}{5^6} = 1 - \frac{1}{64} + \frac{1}{729} - \frac{1}{4096} + \frac{1}{15625} \approx 0.9855$$